Exercise (From HOTS): This is one version of the famous "Let's Make a Deal" or "Monty Hall" game show problem. There are three doors. You know that there is a prize behind one of them, and nothing behind the other two. The game show host tells you that you shall receive whatever is behind the door of your choice. However, before you choose, he promises that rather than immediately opening the door of your choice to reveal its contents, he will open one of the other two doors to reveal that it is empty. He will then give you the option to switch your choice. You may assume that the host is totally impartial – not malicious in any way.

You choose Door 3. He opens Door 2 and reveals that it is empty. You now know that the prize lies behind either Door 3 or Door 1. Should you switch you choice to Door 1?

$B_i = \{\text{prize behind door } i\}$

$H_i = \{\text{host opens door } i\}$

$P(B_1 | H_2) = \text{prob. wins if switch}$

$P(B_3 | H_2) = \text{prob. wins if don't switch} = 1/3 = P(B_3)$

$$P(B_1 | H_2) = \frac{P(H_2|B_1)P(B_1)}{P(H_2|B_1)P(B_1) + P(H_2|B_2)P(B_2) + P(H_2|B_3)P(B_3)}$$

Using Bayes' Rule

$$= \frac{(1)(1/3)}{(1)(1/3) + (0)(1/3) + (1/2)(1/3)} = \frac{2}{3}$$

Heuristic

Suppose at outset, host offers two choices:

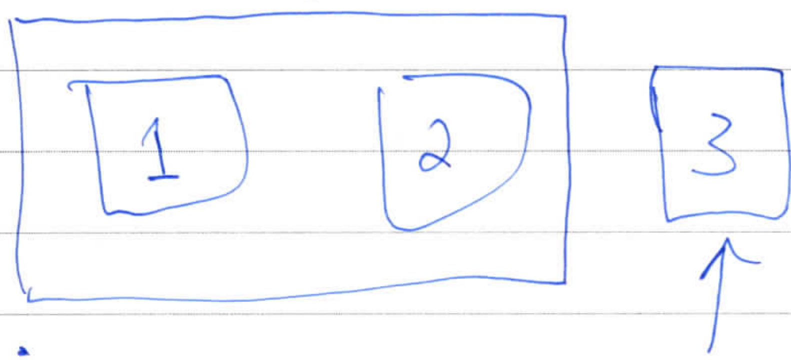
(1) Choose two doors, and win if prize behind either

(2) Choose one door, and win if it has the prize

No switching.

Under original problem:

When switch, you are "choosing two doors"



Or:

Imagine there are 100 doors.

Contestant chooses 1

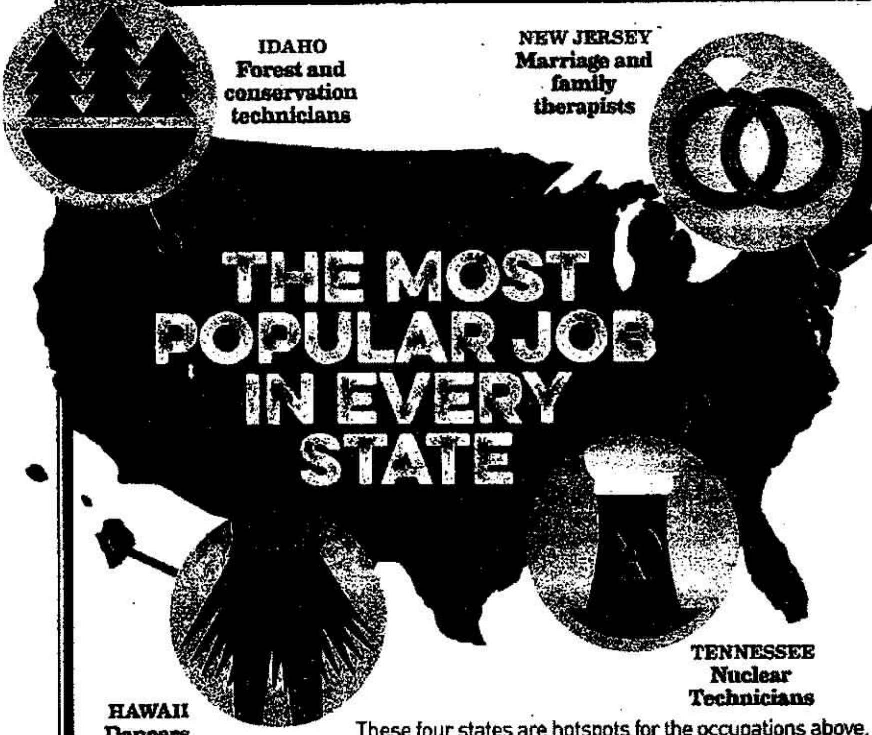
Host opens 98 doors showing no prize

Should you switch

Parade Picks

Edited by Vi-An Nguyen / LIKE US AT FACEBOOK.COM/PARADEMAG

THE MOST POPULAR JOB IN EVERY STATE



These four states are hotspots for the occupations above, according to the U.S. Bureau of Labor Statistics.* You'll never guess which job is unusually popular in Alaska! Check out a chart of all 50 states' top jobs at **parade.com/labor.**

*The jobs have the highest relative concentration in their respective states.

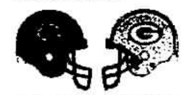
LASSIE TO THE RESCUE

For National Preparedness Month, America's favorite heroic dog is teaming up with Save the Children to spread the word about the importance of emergency kits for kids. From medical info to recent photos, learn what should go in every kit—and see how Lassie saves the day when she drops by the *Parade* offices—at **parade.com/lassie.**

CLASH OF THE TITANS

As the NFL season kicks off this week, we asked the sportscasters of NBC's *Sunday Night Football* and *Football Night in America* which big games they're most excited to watch. Share yours at parade .com/NFL.

AL MICHAELS and CRIS COLLINSWORTH: San Francisco at Denver (Oct. 19) "This is a possible Super Bowl preview," Michaels says. "These are two of last year's best teams, with star QBs—one at the end of a great career [Peyton Manning] and one at the start [Colin Kaepernick]," Collinsworth adds.



BOB COSTAS: Chicago at Green Bay (Nov. 9) "All of the matchups look good. But the Bears and the Packers are the most-played rivalry in the league," Costas says.

MICHELE TAFOYA and HINES WARD: Seattle at San Francisco (Thanksgiving) "Hands down, it's the NFC Championship Game rematch—a fierce rivalry with exciting quarterbacks and loads of fascinating personalities," Tafoya says. "These teams have genuine dislike but the utmost respect for each other," Ward says.

Ask Marilyn

By Marilyn vos Savant

Say that 1 million scratch-off cards are sold. Under the silver-coated area, one card reads, "You win a date with the man or woman of your dreams." All the other cards read, "You lose." After buying a card for myself, I observe 999,998 other contestants scratch off their cards to read, "You lose."

Words We Need

izzat (interjection) reflexive comment made upon spotting a celebrity

A 999,999th contestant, his unscratched card still in his back pocket, is playing Frisbee with his dog. Should I offer him a lot of money to swap scratch-off cards with me?

—B. H., San Jose, California

No, because your chance of winning (now 50-50) wouldn't improve. Instead, offer him a lot of money to *sell* his card to you.

Numbrix

Complete 1 to 81 so the numbers follow a horizontal or vertical path—no diagonals.

3		5		43		49		51
9								77
11								75
21								69
29		31		33		63		67

PHOTOS: CLOCKWISE FROM TOP: COURTESY OF THE NFL (4); RACHAEL SHANE; ILLUSTRATION: SCRIPT & SEAL

Parade Magazine, August 31, 2014

Exercise: It is difficult to conduct sample surveys on sensitive issues because many people will not answer questions if the answers might be embarrassing or incriminating. *Randomized Response* is an effective way to guarantee anonymity while collecting information on topics such as student cheating or tax evasion. (Warner, *Journal of the American Statistical Association*, 1965).

For example, to ask a sample of students whether they have ever used illegal drugs, have each student toss a fair coin *in private*. If the coin lands **Heads**, they are to tell the **truth**. If the coin lands **Tails** they are to answer **"Yes"** (regardless of the truth). Only the student knows whether the "embarrassing" answer ("Yes") reflects the truth or just the coin toss.

Explain how the randomized response method can be used to estimate the probability that a student has used illegal drugs.

$A = \{$ student responds "Yes" $\}$

$B = \{$ coin lands "Heads" $\}$

Want $P(A|B)$ = prob. randomly chosen student responds "Yes" when asked to respond honestly

= proportion of students who use illegal drugs

What do we know? (At least approximately)

We observe the proportion of students who respond "Yes." This is an estimate of $P(A)$.

By LFTP,

$$P(A) = P(A|B)P(B) + P(A|B^c)P(B^c)$$
$$= P(A|B)\left(\tfrac{1}{2}\right) + (1)\left(\tfrac{1}{2}\right)$$

So, $P(A|B) = 2P(A) - 1$

e.g. If 60 out of 100 respond "Yes", then estimate $P(A) \approx 0.6$ and $P(A|B) \approx 0.2$

Independent Events

A pair of events A, B are defined to be **independent** if $P(A \cap B) = P(A)P(B)$.

$P(A|B) = P(A)$ $\quad$ $P(B|A) = P(B)$

A sequence of events $A_1, A_2, \ldots, A_n$ are independent if for **every** collection of indices

$$\{i_1, i_2, \ldots, i_k\} \subseteq \{1, 2, \ldots, n\}$$

for $k \leq n$ it holds that

$$P(A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_k}) = \prod_{j=1}^{k} P(A_{i_j})$$

Comment: Note that for a sequence of events $A_1, A_2, \ldots, A_n$ it is **not** sufficient that

$$P(A_1 \cap A_2 \cap \cdots A_n) = \prod_{i=1}^{n} P(A_i)$$

in order that $A_1, A_2, \ldots, A_n$ be independent.

But, of course, if $A_1, A_2, \ldots, A_n$ are **assumed** to be independent, then it holds that

$$P(A_1 \cap A_2 \cap \cdots A_n) = \prod_{i=1}^{n} P(A_i)$$

This is the most commonly used application of independent events.

[redacted]

'layer: **15sigma22pa**

KNOWANDSHOW5

CLICK HERE for Know And Show GLOSSARY of TERMS

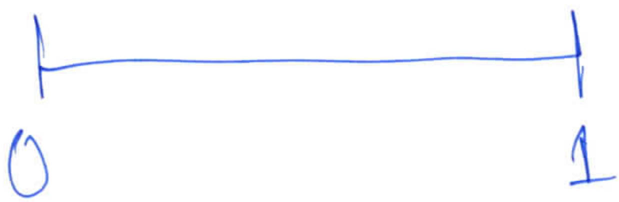
Category 6: Chances Are

Points Earned: 0

Three out of 45 games that Arlene bowls, she scores a minimum of 280 points. How would you describe her chances that on her next 18 games she will bowl at least one game with a score of 280 points or higher?

click answer

Good A
Very Poor B
Poor C
Very Good D



$A_i = \{\text{she scores at least 280 on game } i\}$
$i = 1, 2, \ldots, 18$

$P(\underline{\text{at least one}} \text{ game of at least 280})$
$= 1 - P(\text{no games of at least 280})$
$= 1 - P(A_1^c \cap A_2^c \cap \cdots \cap A_{18}^c)$
$= 1 - P(A_1^c)P(A_2^c)\cdots P(A_{18}^c)$ — assuming games are independent

$= 1 - \left(\frac{14}{15}\right)^{18} \approx 0.71$ "very good"

Rule of thumb: In a sequence of n independent trials, with a prob. of "success" of $1/n$ on each, the prob. of at least one success is approx. $2/3$

$$1 - \left(\frac{n-1}{n}\right)^n \approx \frac{2}{3}$$

$\lim_{n \uparrow \infty}$ is $1 - e^{-1}$

Exercise (From HOTS): If you like gambling and you like betting on the outcome of sports matches, then you may like the "parlay card." A parlay card lets you bet on the outcomes of more than one match. In order to win a parlay bet, you must be correct on each of the matches you bet upon. Parlay cards offer big payoffs if you are right on every match.

Suppose that your bookie will give you 10-to-1 odds for a parlay bet that covers four sports matches. Should you take the bet?

"10 to 1 odds" $\Rightarrow$ $P(\text{lose bet})/P(\text{win bet}) = 10$

Let $p = P(\text{bettor wins})$, then $\frac{1-p}{p} = 10$ when $p = \frac{1}{11}$

If I assume the chance of me choosing correctly on each match is $1/2$ and the matches are independent, then

$$P(\text{bettor wins}) = \left(\frac{1}{2}\right)^4 = \frac{1}{16}$$

So, since $1/16 < 1/11$, should not take bet

Conditional Independence

Suppose that events A_1 and A_2 are not necessarily independent, but there is another event B such that

$$P(A_1 \cap A_2 | B) = P(A_1 | B) P(A_2 | B)$$

Then we say that A_1 and A_2 are **independent conditional on** B.

Example: Imagine that there are three coins: One is a regular two-sided coin, another has "Heads" on both sides, and another has "Tails" on both sides. One of three coins is chosen at random, and then flipped twice. Let A_i be the event that flip i comes up "Heads."

$P(A_2 | A_1) \neq P(A_2)$

Then, A_1 and A_2 are dependent, but if I define B_j to be the event that the chosen coin has j sides with "Heads," (with $j = 0, 1, 2$), then A_1 and A_2 are independent conditional on any one of the B_j.

Exercise: A bank classifies customers as either a good or bad credit risk. On the basis of extensive historical data, the bank has observed that 1% of good credit risks and 10% of bad credit risks overdraw their account in any given month. A new customer opens a checking account at the bank. On the basis of a check with the credit bureau, the bank believes that there is a 70% chance that the customer will turn out to be a good credit risk.

Suppose that this customer's account is overdrawn in the first month. How does this alter the bank's opinion of this customer's creditworthiness? What if the customer overdraws the account again in the second month?

$B = \{$ person is a good credit risk $\}$
$A_i = \{$ person overdraws in month i $\}$ $i = 1, 2$

$P(B) = 0.70$ "prior probability"
$P(A_i | B) = 0.01$ $P(A_i | B^c) = 0.10$

First question asks for $P(B | A_1)$
Use Bayes' Rule, $P(B | A_1) = \underline{0.19}$

Now get more data: customer overdraws in month 2

Redo Bayes' Rule with "prior prob" of 0.19 and find Prob. is good credit risk is $0.02 = P(B \mid A_1 \cap A_2)$

Imagine instead the data is observed after two months, i.e. the "data" is $A_1 \cap A_2$

Still $P(B) = 0.70$.

$$P(B \mid \underline{A_1 \cap A_2}) = \frac{P(A_1 \cap A_2 \mid B)\,P(B)}{P(A_1 \cap A_2 \mid B)\,P(B) + P(A_1 \cap A_2 \mid B^c)\,P(B^c)}$$

using Bayes' Rule

$$= \frac{P(A_1 \mid B)\,P(A_2 \mid B)\,P(B)}{P(A_1 \mid B)\,P(A_2 \mid B)\,P(B) + P(A_1 \mid B^c)\,P(A_2 \mid B^c)\,P(B^c)}$$

<u>assuming</u> A_1, A_2 independent conditional on B

which implies A_1, A_2 " " " B^c

$= 0.02$

<u>Bayesian updating</u> is assuming conditional independence

Part 2: Random Variables

A **random variable (RV)** is a function that assigns a real value to each outcome $\omega \in \Omega$.

We almost always use capital letters late in the alphabet such as U, V, X, Y, and Z to denote random variables.

Example: Suppose that X equals the number of tosses of a fair coin in order to see the first "Heads."

Then, one could write

$$\Omega = \{\text{H}, \text{TH}, \text{TTH}, \text{TTTH}, \ldots\}$$

and then would define $X : \Omega \to \mathbb{R}$ as

$$\begin{aligned} X(\text{H}) &= 1 \\ X(\text{TH}) &= 2 \\ X(\text{TTH}) &= 3 \\ &\vdots \end{aligned}$$

When we write "$P(X \le c)$," for instance, it should be understood is that this means

$$P(\{\omega\colon X(\omega) \le c\})\,.$$

When it is written in this way, it is clear that

$$\{\omega\colon X(\omega) \le c\}$$

is a subset of Ω.

Exercise: Is it necessarily the case that

$$P(\{\omega\colon X(\omega) \le c\})$$

is defined?

No. In order for it to be defined, need $\{\omega : X(\omega) \le c\} \in \mathcal{F}$.

If $\{\omega : X(\omega) \le c\} \in \mathcal{F}$ for all c, then "X is measurable"

The **cumulative distribution function (CDF)** for a random variable X is

$$F_X(c) = P(X \le c)$$

Exercise: Two fair coins are tossed and the outcome is observed. Before the coins are tossed, we are given a choice of the following payoffs:

Payoff 1: Win \$1 for each head.
Lose \$3 for getting two tails.

Payoff 2: Win \$1 if the coins are different.
Win \$2 if both coins turn up tails.
Lose \$3 if both coins turn up heads.

Which payoff should we choose? Write the CDFs of the random variables.

$\Omega = \{HH, HT, TH, TT\}$ ELO

$X =$ payoff under #1
$Y =$ " " #2

$X(HH) = 2$ $\quad Y(HH) = -3$
$X(HT) = 1$ $\quad Y(HT) = 1$
$X(TH) = 1$ $\quad Y(TH) = 1$
$X(TT) = -3$ $\quad Y(TT) = 2$

Note that $X \neq Y$, i.e. they are <u>not</u> the same function on Ω
($X = Y$ means $X(\omega) = Y(\omega)$ for all $\omega \in \Omega$)

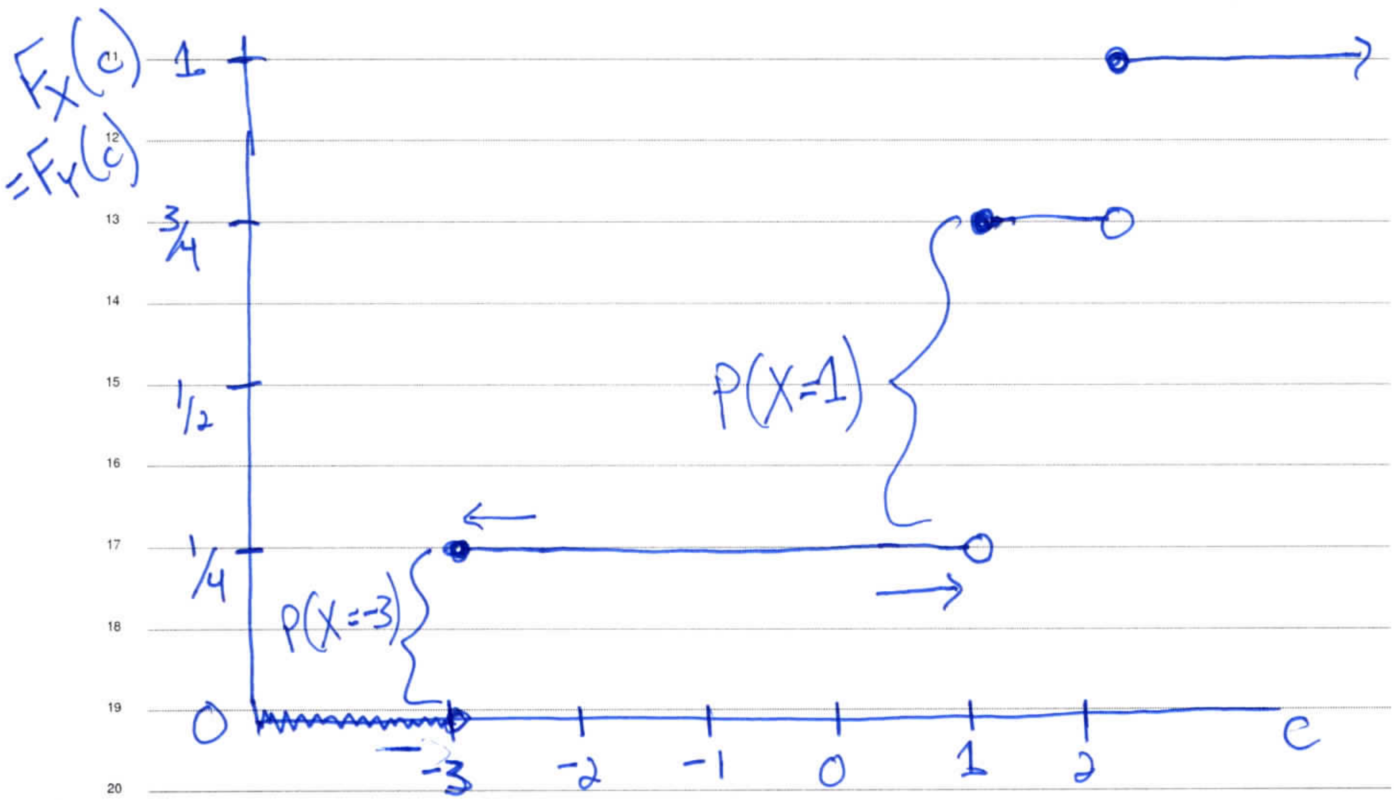
But, X, Y have the same distribution, i.e.

$$P(X \in A) = P(Y \in A) \quad \text{for any } A \subseteq \mathbb{R}$$

(for which the prob. is defined)

$$\boxed{\begin{array}{l}\text{Comment: If I write } X+Y \text{, I mean defining } Z : \Omega \to \mathbb{R} \text{ such that} \\ \qquad Z(\omega) = X(\omega) + Y(\omega) \quad \text{for all } \omega \in \Omega\end{array}}$$

What is the CDF of X (and of Y)?



Properties of any CDF

The CDF F_X must satisfy the following properties:

1. $F_X(c)$ is nondecreasing as a function of c.
2. $\lim\limits_{c \nearrow \infty} F_X(c) = 1$
3. $\lim\limits_{c \searrow -\infty} F_X(c) = 0$
4. F_X is **right-continuous,** i.e. $\lim\limits_{y \searrow c} F_X(y) = F_X(c)$
5. $P(X > c) = 1 - F_X(c)$.
6. $P(X = c) = F_X(c) - \lim\limits_{y \nearrow c} F_X(y) =$ size of jump in CDF at c
7. $P(a < X \leq b) = F_X(b) - F_X(a)$
8. The CDF uniquely determines the distribution. In other words, if two random variables have the same CDF, they have the same distribution.

Crucial consequence of Property 8: If we derive the CDF of a random variable, and find that it matches the CDF of some "known" distribution, then can claim that the random variable has that particular distribution.

Basics of Discrete Random Variables

$(X-k)^+$

Distributions of random variables are divided into three groups: discrete, continuous, and mixed.

Discrete distributions give positive probability to a countable collection of values. Often this collection is a subset of the nonnegative integers, but that is not part of the definition.

A discrete random variable is characterized by its **probability mass function (pmf)**, denoted f_X:

$$f_X(k) = P(X = k)$$

The **expected value** of a discrete random variable is then defined naturally as

$$E(X) = \sum_{k:\, f_X(k) > 0} k f_X(k)$$

Note: often true that $P(X = E(X)) = 0$

Exercise (From HOTS): You and I are to play a game. You roll a die until a number other than a one appears. When such a number appears for the first time, I pay you the same number of dollars as there are dots on the upturned face of the die, and the game ends. What is the expected payoff to this game?

$X = \#$ on last roll

$P(X=k) = 1/5, \quad k=2,3,4,5,6$

$f_X(2) = f_X(3) = \cdots = f_X(6) = 1/5$

$$E(X) = \sum_{k : f_X(k) > 0} k\, f_X(k)$$

$$= (2)\left(\tfrac{1}{5}\right) + (3)\left(\tfrac{1}{5}\right) + \cdots + (6)\left(\tfrac{1}{5}\right)$$

$$= 4$$

Exercise: Suppose that random variable X has pmf

$$f_X(k) = \frac{1}{k^s \underbrace{z(s)}_{\text{normalizing constant}}}, \quad \text{for } k = 1, 2, \ldots, \text{ and } s > 1$$

where

$$z(s) = \sum_{i=1}^{\infty} \frac{1}{i^s}$$

is the **zeta function.** (This is the **Zeta Distribution**.)

What is $E(X)$ in this case?

$\sum_{k: f_X(k)} f_X(k) = 1$ for any pmf

$$E(X) = \sum_{k=1}^{\infty} k f_X(k) = \sum_{k=1}^{\infty} \left(\frac{1}{k^{s-1}}\right)\left(\frac{1}{z(s)}\right)$$

$$\boxed{\text{Recall: } \sum_{i=1}^{\infty} \frac{1}{i} = \infty, \quad \sum_{i=1}^{\infty} \left(\frac{1}{i}\right)^s < \infty \text{ for } s>1}$$

If $s = 2$,

$$E(X) = \frac{1}{z(2)} \sum_{k=1}^{\infty} \frac{1}{k} = \infty$$

If $s > 2$,

$$E(X) = \frac{1}{z(s)} \sum_{k=1}^{\infty} \frac{1}{k^{s-1}} = \frac{z(s-1)}{z(s)}$$

For $1 < s < 2$, also holds that $E(X) = \infty$

Note: $E(X) = \infty$, even though $P(X = \infty) = 0$